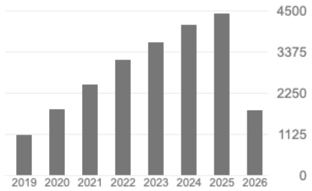
Publications

Summary of Scientific Impact

	All	Since 2021
# Publications	111	85
# Citations	23735	19702
h-index	40	39
i10-index	68	66



per Google Scholar, retreived on June 5th, 2026.

List of Publications

Journal Articles

1. Anders C J, Neumann D, Samek W, Müller K-R and **Lapuschkin S** (2026).
   "Software for Dataset-wide XAI: From Local Explanations to Global Insights with Zennit, CoRelAy, and ViRelAy".
   In: *PLoS ONE* 21(1):e0336683.
   https://github.com/chr5tphr/zennit | *https://github.com/virelay/corelay* | *https://github.com/virelay/virelay*
2. Cantú E D, Wittmann R K, Abdeen O, Wagner P, Samek W, Baier M and **Lapuschkin S** (2026).
   "Deep Learning-based Multi Project InP Wafer Simulation towards Unsupervised Surface Defect Detection".
   In: *Journal of the Franklin Institute* 363(2):108316
3. Yolcu G Ü, Weckbecker M, Wiegand T, Samek W and **Lapuschkin S** (2025).
   "Sparse, Efficient and Explainable Data Attribution with DualXDA".
   In: *Transactions on Machine Learning Research* qfx81N884A.
   https://github.com/gumityolcu/DualXDA
4. Dreyer M, Berend J, Labarta T, Vielhaben J, Wiegand T, **Lapuschkin S** and Samek W (2025).
   "Mechanistic Understanding and Validation of Large AI Models with SemanticLens".
   In: *Nature Machine Intelligence* 1–14.
   https://github.com/jim-berend/semanticlens | *Demo: https://semanticlens.hhi-research-insights.eu*
5. Pahde F, Wiegand T, **Lapuschkin S** and Samek W (2025).
   "Ensuring Medical AI Safety: Explainable AI-Driven Detection and Mitigation of Spurious Model Behavior and Associated Data".
   In: *Machine Learning* 114(9):206.
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6. Ma J, Weicken E, Pahde F, Weitz K, **Lapuschkin S**, Samek W and Wiegand T (2025).
   "Künstliche Intelligenz auf dem Prüfstand: Anforderungen, Qualitätskriterien und Prüfwerkzeuge für medizinische Anwendungen [*Artificial intelligence under scrutiny: requirements, quality criteria, and testing tools for medical applications*]".
   In: *Bundesgesundheitsblatt – Gesundheitsforschung – Gesundheitsschutz* 68:915-923
7. Storås A M, Dreyer M, Pahde F, **Lapuschkin S**, Samek W, Halvorsen P, de Lange T, Mori Y, Hann A, Berzin T M, Parasa S and Riegler M A (2025).
   "Exploring the Clinical Value of Concept-based AI Explanations in Gastrointestinal Disease Detection".
   In: *Scientific Reports* 15(1):28860.
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8. Weber L, Berend J, Weckbecker M, Binder A, Wiegand T, Samek W and **Lapuschkin S** (2025).
   "Efficient and Flexible Neural Network Training through Layer-wise Feedback Propagation".
   In: *Transactions on Machine Learning Research* 9oToxYVOSW.
   https://github.com/leanderweber/layerwise-feedback-propagation
9. Hedström A, Bommer P L, Burns T F, **Lapuschkin S**, Samek W and Höhne M-C M (2025).
   "Evaluating Interpretable Methods via Geometric Alignment of Functional Distortions".
   In: *Transactions on Machine Learning Research* ukLxqA8zXj.
   https://github.com/annahedstroem/GEF/ | *TMLR Survey Certification*

10. Bley F, **Lapuschkin S**, Samek W and Montavon G (2025).
"Explaining Predictive Uncertainty by Exposing Second-Order Effects".
In: *Pattern Recognition* 160:111171.
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11. Vielhaben J, **Lapuschkin S**, Montavon G and Samek W (2024).
"Explainable AI for Time Series via Virtual Inspection Layers".
In: *Pattern Recognition* 150:110309.
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12. Becker S, Vielhaben J, Ackermann M, Müller K-R, **Lapuschkin S** and Samek W (2024).
"AudioMNIST: Exploring Explainable Artificial Intelligence for Audio Analysis on a Simple Benchmark".
In: *Journal of the Franklin Institute* 361(1):418–428.
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13. Achtibat R, Dreyer M, Eisenbraun I, Bosse S, Wiegand T, Samek W and **Lapuschkin S** (2023).
"From attribution maps to human-understandable explanations through Concept Relevance Propagation".
In: *Nature Machine Intelligence* 5(9):1006–1019.
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14. Hedström A, Bommer P, Wickstrøm K K, Samek W, **Lapuschkin S** and Höhne M-C M (2023).
"The Meta-Evaluation Problem in Explainable AI: Identifying Reliable Estimators with `MetaQuantus`".
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15. Weber L, **Lapuschkin S**, Binder A and Samek W (2023).
"Beyond Explaining: Opportunities and Challenges of XAI-Based Model Improvement".
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16. Hedström A, Weber L, Krakowczyk D G, Bareeva D, Motzkus F, Samek W, **Lapuschkin S** and Höhne M-C M (2023).
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17. Hofmann S M, Beyer F, **Lapuschkin S**, Goltermann O, Loeffler M, Müller K-R, Villringer A, Samek W and Witte A V (2022).
"Towards the Interpretability of Deep Learning Models for Multi-modal Neuroimaging: Finding Structural Changes of the Ageing Brain".
In: *NeuroImage* 261:119504

18. Ma J, Schneider L, **Lapuschkin S**, Achtibat R, Duchrau M, Krois J, Schwendicke F and Samek W (2022).
"Towards Trustworthy AI in Dentistry".
In: *Journal of Dental Research* 00220345221106086

19. Rieckmann A, Dworzynski P, Arras L, **Lapuschkin S**, Samek W, Onyebuchi A A, Rod N H, Ekstrøm C T (2022).
"Causes of Outcome Learning: A Causal Inference-inspired Machine Learning Approach to Disentangling Common Combinations of Potential Causes of a Health Outcome".
In: *International Journal of Epidemiology* dyac078.
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20. Slijepcevic D, Horst F, **Lapuschkin S**, Horsak B, Raberger A-M, Kranzl A, Samek W, Breiteneder C, Schöllhorn W I and Zeppelzauer M (2022).
"Explaining Machine Learning Models for Clinical Gait Analysis".
In: *ACM Transactions on Computing for Healthcare* 3(2):14:1–27.
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21. Anders C J, Weber L, Neumann D, Samek W, Müller K-R and **Lapuschkin S** (2022).
"Finding and Removing Clever Hans: Using Explanation Methods to Debug and Improve Deep Models".
In: *Information Fusion* 77:261–295

22. Sun J, **Lapuschkin S**, Samek W and Binder A (2022).
"Explain and Improve: LRP-inference Fine-tuning for Image Captioning Models".
In: *Information Fusion* 77:233–246

23. Samek W, Montavon G, **Lapuschkin S**, Anders C J, and Müller K-R (2021).
"Explaining Deep Neural Networks and Beyond: A Review of Methods and Applications".
In: *Proceedings of the IEEE* 109(3):247–278

24. Yeom S-K, Seegerer P, **Lapuschkin S**, Binder A, Wiedemann S, Müller K-R and Samek W (2021).
"Pruning by Explaining: A Novel Criterion for Deep Neural Network Pruning".
In: *Pattern Recognition* 115:107899.
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25. Aeles J, Horst F, **Lapuschkin S**, Lacourpaille L, and Hug F (2021).
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26. Horst F, Slijepcevic D, Zeppelzauer M, Raberger AM, **Lapuschkin S**, Samek W, Schöllhorn WI, Breiteneder C, and Horsak B (2020).
"Explaining Automated Gender Classification of Human Gait".
In: *Gait & Posture* 81(S1):159–160
27. Hägele M, Seegerer P, **Lapuschkin S**, Bockmayr M, Samek W, Klauschen F, Müller K-R and Binder A (2020).
"Resolving Challenges in Deep Learning-based Analyses of Histopathological Images using Explanation Methods".
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28. Alber M, **Lapuschkin S**, Seegerer P, Hägele M, Schütt K T, Montavon G, Samek W, Müller K-R, Dähne S and Kindermans P-J (2019).
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In: *Journal of Machine Learning Research* 20(93):1–8.
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29. **Lapuschkin S**, Wäldchen S, Binder A, Montavon G, Samek W and Müller K-R (2019).
"Unmasking Clever Hans Predictors and Assessing what Machines Really Learn".
In: *Nature Communications* 10:1069
30. Horst F, **Lapuschkin S**, Samek W, Müller K-R and Schöllhorn W I (2019).
"Explaining the Unique Nature of Individual Gait Patterns with Deep Learning".
In: *Scientific Reports* 9:2391.
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31. Montavon G, **Lapuschkin S**, Binder A, Samek W and Müller K-R (2017).
"Explaining NonLinear Classification Decisions with Deep Taylor Decomposition".
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Pattern Recognition Best Paper Award and Pattern Recognition Medal winner
32. Samek W, Binder A, Montavon G, **Lapuschkin S**, and Müller K-R (2017).
"Evaluating the Visualization of what a Deep Neural Network has Learned".
In: *IEEE Transactions of Neural Networks and Learning Systems*
33. Sturm I, **Lapuschkin S**, Samek W and Müller K-R (2016).
"Interpretable Deep Neural Networks for Single-Trial EEG Classification".
In: *Journal of Neuroscience Methods* 274:141–145
34. **Lapuschkin S**, Binder A, Montavon G, Müller K-R and Samek W (2016).
"The Layer-wise Relevance Propagation Toolbox for Artificial Neural Networks".
In: *Journal of Machine Learning Research* 17(114):1–5.
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35. **Bach S**, Binder A, Montavon G, Klauschen F, Müller K-R and Samek W (2015).
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In: *PLoS ONE* 10(7):e0130140

Contributions to Conference Proceedings and Workshops

1. Ostermann S, Gurgurov D, Baeumel T, Hedderich M A, **Lapuschkin S**, Samek W and Schmitt V (2026).
"From Weights to Activations: Is Steering the Next Frontier of Adaptation?".
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2. Komorowski P, Golimblevskaia E, Achtibat R, Wiegand T, **Lapuschkin S** and Samek W (2026).
"Attribution-Guided Decoding".
In: *Proceedings of the International Conference on Learning Representations (ICLR)* .
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3. Kahardipraja P, Achtibat R, Wiegand T, Samek W and **Lapuschkin S** (2025).
"The Atlas of In-Context Learning: How Attention Heads Shape In-Context Retrieval Augmentation".
In: *Advances in Neural Information Processing Systems (NeurIPS)* .
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4. Bareeva D, Höhne M M C, Warnecke A, Pirch L, Müller K-R, Rieck K, **Lapuschkin S** and Bykov K (2025). "Manipulating Feature Visualizations with Gradient Slingshots".
In: *Advances in Neural Information Processing Systems* (*NeurIPS*) .
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5. Sandmann E, **Lapuschkin S** and Samek W (2025).
"The Algorithm Is Not the Behavior: Learned Priors Override Look-Ahead in a Chess-Playing Neural Network".
In: *NeurIPS 2025 Workshop on Mechanistic Interpretability* .
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6. Golimblevskaia E, Jain A, Puri B, Ibrahim A, Samek W and **Lapuschkin S** (2025).
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In: *Proceedings of the 8th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP* (*EMNLP*) .
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7. Labarta T, Hoang N, Weitz K, Samek W, **Lapuschkin S** and Weber L (2025).
"See What I Mean? CUE: A Cognitive Model of Understanding Explanations".
In: *Proceedings of the IJCAI Workshops 2025: XAI Workshop* .
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8. Puri B, Jain A, Golimblevskaia E, Kahardipraja P, Wiegand T, Samek W and **Lapuschkin S** (2025).
"FADE: Why Bad Descriptions Happen to Good Features".
In: *Findings of the Association for Computational Linguistics* (*ACL*) 17138–17160.
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9. Naujoks J, Krasowski A, Weckbecker M, Yolcu G Ü, Wiegand T, **Lapuschkin S**, Samek W and Klausen R P (2025).
"Leveraging Influence Functions for Resampling Data in Physics-Informed Neural Networks".
In: *Proceedings of the 3rd XAI World Conference* 383–395.
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10. Erogullari E, **Lapuschkin S**, Samek W and Pahde F (2025).
"Post-Hoc Concept Disentanglement: From Correlated to Isolated Concept Representations".
In: *Proceedings of the 3rd XAI World Conference* 68–89.
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11. Joseph S, Suresh P, Hufe L, Stevinson E, Graham R, Vadi Y, Bzdok D, **Lapuschkin S**, Sharkey L and Richards A (2025).
"Prisma: An Open Source Toolkit for Mechanistic Interpretability in Vision and Video".
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12. Pahde F, Dreyer M, Weckbecker M, Weber L, Anders C J, Wiegand T, Samek W and **Lapuschkin S** (2025).
"Navigating Neural Space: Revisiting Concept Activation Vectors to Overcome Directional Divergence".
In: *Proceedings of the International Conference on Learning Representations* (*ICLR*) .
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13. Bareeva D, Yolcu GÜ, Hedström A, Wiegand T, Samek W and **Lapuschkin S** (2024).
"Quanda: An Interpretability Toolkit for Training Data Attribution Evaluation and Beyond".
In: *NeurIPS 2024 Workshop on Attributing Model Behavior at Scale* (*ATTRIB 2024*) .
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14. Naujoks J R, Krasowski A, Weckbecker M, Wiegand T, **Lapuschkin S**, Samek W and Klausen R P (2024).
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Reproducibility Badge Winner
15. Kopf L, Bommer P L, Hedström A, **Lapuschkin S**, Höhne M M-C and Bykov K (2024).
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In: *Advances in Neural Information Processing Systems* (*NeurIPS*) 34656–34685. (*OpenReview*)
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16. Nobis G, Springenberg M, Aversa M, Detzel M, Daems R, Murray-Smith R, Nakajima S, **Lapuschkin S**, Ermon S, Birdal T, Opper M, Knochenhauer C, Oala L and Samek W (2024).
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17. Mekala R R, Pahde F, Baur S, Chandrashekar S, Diep M, Wenzel M A, Wisotzky E L, Yolcu G Ü, **Lapuschkin S**, Ma J, Eisert P, Lindvall M, Porter A and Samek W (2024).
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18. Achtibat R, Hatefi S M V, Dreyer M, Jain A, Wiegand T, **Lapuschkin S**, Samek W (2024).
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19. Hatefi S M V, Dreyer M, Achtibat R, Wiegand T, Samek W and **Lapuschkin S** (2024).
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In: *Proceedings of the European Conference on Computer Vision* (*ECCV*) *Workshops* 152–169. (*Green Open Access*)
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20. Hedström A, Weber L, **Lapuschkin S**, Höhne M M-C (2024).
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In: *Proceedings of the 2nd XAI World Conference* 403–420. (*Green Open Access*)
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21. Tinauer C, Damulina A, Sackl M, Soellradl M, Achtibat R, Dreyer M, Pahde F, **Lapuschkin S**, Schmidt R, Ropele S, Samek W, Langkammer C (2024).
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22. Dreyer M, Purelku E, Vielhaben J, Samek W, **Lapuschkin S** (2024).
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23. Bareeva D, Dreyer M, Pahde F, Samek W and **Lapuschkin S** (2024).
"Reactive Model Correction: Mitigating Harm to Task-Relevant Features via Conditional Bias Suppression".
In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*) *Workshops* 3532–3541.
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24. Dreyer M, Achtibat R, Samek W and **Lapuschkin S** (2024).
"Understanding the (Extra-)Ordinary: Validating Deep Model Decisions with Prototypical Concept-based Explanations".
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25. Dreyer M, Pahde F, Anders C J, Samek W and **Lapuschkin S** (2024).
"From Hope to Safety: Unlearning Biases of Deep Models via Gradient Penalization in Latent Space".
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26. Dawoud K, Samek W, Eisert P, **Lapuschkin S** and Bosse S (2023).
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In: *Proceedings of the IEEE International Conference on Data Mining* (*ICDM*) 912–921. (*Green Open Access*)

27. Frommholz A, Seipel F, **Lapuschkin S**, Samek W and Vielhaben J (2023).
"XAI-based Comparison of Audio Event Classifiers with different Input Representations".
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28. Hedström A, Weber L, **Lapuschkin S** and Höhne M M-C (2023).
"Sanity Checks Revisited: An Exploration to Repair the Model Parameter Randomisation Test".
In: *NeurIPS 2023 Workshop on XAIX* (*XAI in Action: Past, Present, and Future Applications*) (vVpefYmnsG)

29. Pahde F, Dreyer M, Samek W and **Lapuschkin S** (2023).
"Reveal to Revise: An Explainable AI Life Cycle for Iterative Bias Correction of Deep Models".
In: *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention* (*MICCAI*) 596–606. (*Green Open Access*)
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"Shortcomings of Top-Down Randomization-Based Sanity Checks for Evaluations of Deep Neural Network Explanations".
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31. Dreyer M, Achtibat R, Wiegand T, Samek W and **Lapuschkin S** (2023).
"Revealing Hidden Context Bias in Segmentation and Object Detection through Concept-specific Explanations".
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32. Pahde F, Yolcu GÜ, Binder A, Samek W and **Lapuschkin S** (2023).
"Optimizing Explanations by Network Canonization and Hyperparameter Search".
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33. Krakowczyk D G, Prasse P, Reich D R, **Lapuschkin S**, Scheffer T, Jäger L A (2023).
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34. Krakowczyk D G, Reich D R, Prasse P, **Lapuschkin S**, Jäger L A and Scheffer T (2022).
"Selection of XAI Methods Matters: Evaluation of Feature Attribution Methods for Oculomotoric Biometric Identification".
In: *NeurIPS 2022 Workshop on Gaze Meets ML* (GOLdDAP2AtI)
35. Motzkus F, Weber L and **Lapuschkin S** (2022).
"Measurably Stronger Explanation Reliability via Model Canonization".
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36. Ede S, Baghdadlian S, Weber L, Nguyen A, Zanca D, Samek W and **Lapuschkin S** (2022).
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37. Sun J, **Lapuschkin S**, Samek W, Zhao Y, Cheung N-M and Binder A (2021).
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In: *Proceedings of the 25th International Conference on Pattern Recognition* (*ICPR*) 7609–7616
38. Goh G S W, **Lapuschkin S**, Weber L, Samek W and Binder A (2021).
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"Towards Best Practice in Explaining Neural Network Decisions with LRP".
In: *Proceedings of the IEEE International Joint Conference on Neural Networks* (*IJCNN*) 1-7
40. Sun J, **Lapuschkin S**, Samek W and Binder A (2020).
"Understanding Image Captioning Models beyond Visualizing Attention".
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41. Anders C J, Neumann D, Marinč T, Samek W, Müller K-R and **Lapuschkin S** (2020).
"XAI for Analyzing and Unlearning Spurious Correlations in ImageNet".
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42. Sun J, **Lapuschkin S**, Samek W, Zhao Y, Cheung N-M and Binder A (2020).
"Explain and Improve: Cross-Domain-Few-Shot-Learning Using Explanations".
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43. Alber M, **Lapuschkin S**, Seegerer P, Hägele M, Schütt K T, Montavon G, Samek W, Müller K-R, Dähne S and Kindermans P-J (2018).
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44. **Lapuschkin S**, Binder A, Müller K-R and Samek W (2017).
"Understanding and Comparing Deep Neural Networks for Age and Gender Classification".
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45. Srinivasan V, **Lapuschkin S**, Hellge C, Müller K-R and Samek W (2017).
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46. **Bach S**, Binder A, Müller K-R and Samek W (2016).
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